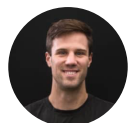


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5 ways to practically apply velocity based training (VBT)

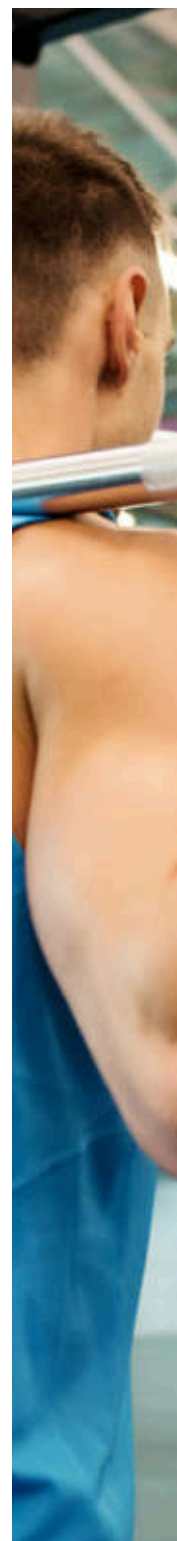
Integrating velocity tracking in your training doesn't have to be difficult! Here are five of the most effective ways to use VBT in the gym



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Co-founder

A common misconception amongst coaches and lifters is that in order to begin using VBT you must completely change the way you approach training.

The reality is much more simple: velocity is just an extra piece of data



we can collect in the gym, not a completely different training philosophy.

It is very similar to how GPS data is now a ubiquitous performance monitoring tool for field and endurance sports. GPS data allows coaches to closely monitor running workload for their athletes in real time. This has revolutionised how coaches prescribe workloads, letting them set highly individualised training to ensure each athlete in their team will perform at their best.

The fundamental rules of training for field and endurance sports didn't get thrown out the window when GPS became widely adopted! Coaches still use the same types of energy system training: sprint work, fartlek sessions, captains runs or small sided games.

Ubiquitous and seamless velocity tracking can provide the same fantastic opportunity for lifting.

Using VBT doesn't change our fundamental approach to training. You still adhere to principals of good strength training using whatever periodisation, load setting or tapering model which works for your goals. Tracking velocity data as you train can assist us dial in our work, and if used correctly will accelerate our progress!

Adapting our programs in response to velocity data can reduce the risk of over and under-training, help improve motor learning processes, set guardrails for rehabilitation, and improve both the creation and execution of strength and power training plans in the gym.

Below I have put together a list of five ways you can start using velocity data to improve your training.

1. Use VBT to train with better intent to move

Intent describes how fast you are trying to move during an exercise.

Lifting with high *intent to move* has well established benefits. Applying high intent on all reps in training (especially compound lifts) leads to greater muscle fibre recruitment, enhanced neural adaptations, improved warm-up effects and overall greater strength and power adaptations. The load may be incredibly heavy and your actual movement velocity might be quite slow, but the act of trying to move with as much intent as possible is the key.

Without velocity tracking we have to guess how much intent is being applied to the bar during a given set. For a coach this is based on how the set looks, and for the lifter how their effort feels.

With velocity data we can instead objectively and precisely determine how much intent is being applied to the bar by monitoring the velocity (measured in m/s) at which it moves.

Velocity data for each load gives lifters real-time feedback on the quality of their training, and offers a new goal to chase: *Velocity PRs*.

Metric provides immediate notifications whenever you hit a new velocity, load, power or rep record for that exercise – the perfect motivator for this.

An
example
of the
record
cards
from the

The simplest way to use this feedback is to just try and max out your velocity on every set, lifting each weight as fast as possible. If you are a coach you can build fantastic competition in your gym by creating a leaderboard for the lifts you want your athletes to put high intent into. This leaderboard could be ranking athletes by who can hit the highest velocity on a 1x bodyweight bench press, or the heaviest weight they can squat above 0.5m/s, or even who has the most powerful 145lbs power clean.

The great thing about velocity is how it can shift our focus away from simply chasing more and more weight on the bar, which can result in overtraining, burnout and reduced performance – the opposite of what we want from our training.

2. Use your best rep velocity to track performance over time

To take your use of velocity to the next level, the Metric app allows you to keep a history of your best rep velocity for the lifts you care about.

You can easily refer to this training history directly within the Metric app with a Pro subscription, aiming to beat your former best velocities and see improvement over time.

As you collect data, the Metric app provides insights into how well you are lifting compared to your last workout or your 6-week average for each exercise and load combination, making it convenient to track your progress.

Ideally, as you lift heavier loads for your working sets, you should also move sub-maximal weights at faster velocities, indicating greater power, rate of force development, and improved technical proficiency across the entire load velocity profile.

An increase of 0.05m/s on the same load over time is correlated with a 5% increase in 1RM for that exercise.

The historical velocity context provided by the Metric app also lets you assess your readiness to train.

While many people try to infer their readiness through sleep tracking, heart rate variability measurements, or subjective self-assessment tools, these methods measure readiness and fatigue status in a general sense. Velocity tracking within the Metric app is a direct measure of your readiness specific to strength training.

Finding that today's velocity is 10% below your recent average for the same load is a reliable indicator that you are in a state of low readiness, and it might be a good day for some back-off work!

The
historical
velocity
performance
within the
Metric
VBT app

3. Use VBT to monitor your range of motion

Range of motion (ROM) is the distance you move the barbell during a repetition. The longer a ROM is, the more work that has to be performed during the rep. ROM is expressed in centimetres (cm).

ROM can be analysed on a rep-by-rep basis and compared to your historical performance. This can be helpful to see if technique changes improve ROM, or if heavier loads result in decreases to ROM.

Metric also offers a *ROM Variance* score for each set. ROM variance is expressed as percent, and represents how much your range of motion deviates from the set average. The closer to 0% the more consistent the ROM was for each rep in the set.

Here are two example sets of Bench Press with their different ROM Variance highlighted. Our goal for most lifters is the excellent consistency as seen in *Set A*, rather than the inconsistency we see in *Set B*.

How you interpret ROM data will depend on your training goals.

If you are lifting for sports performance, hypertrophy, or strength you will want to make sure adding weight to the bar is not coming at the cost of reduced range of motion, and you should ensure to maintain high consistency throughout your sets. It is always nice when the last rep looks the same as the first rep in a set!

Powerlifters can also use ROM data to gauge lift specific technical proficiency, and dial-in their ideal minimum bar travel while still hitting

the requirements for a successful competition lift.

It also interesting to see the relationship between ROM and velocity; a mis groove mid-set might lengthen your ROM but significantly reduce your velocity.

4. Use velocity loss to better understand fatigue across a set

Monitoring how hard your sets are is a great way to auto-regulate your training. The most widely used method for this is RPE (Rating of Perceived Exertion).

Very simply, RPE puts a number value on how close we think we came to our repetition maximum for that weight. Thus RPE-10 is max effort where it isn't possible to do any additional reps, while RPE-6 is an easy set in which you left about 4 reps in the tank. The RPE model was adapted for strength training by powerlifter and coach Mike Tuchscherer back in 2009 in his book [*The Reactive Training Manual*](#).

You can use RPE to program with specific targets in mind (eg. ending a set at RPE-7), or you can use RPE to adjust weight for your top sets on the fly, or a combination.

RPE is very much a subjective measure, but experienced athletes and lifters are pretty accurate at assessing how far from failure they are at the end of each lift.

Velocity based training and RPE training are particularly well suited to work together, due to

the connection between velocity loss, RPE and proximity to failure.

Velocity loss is a measurement of the percentage decline in velocity from the fastest rep of a set to the final rep of a set. Effectively this is an objective measure of what we are trying to gauge with RPE: how close we got to failure.

For any set performed with a high intent to move, there is a correlation between the percentage velocity loss and our proximity to the point of technical failure. For most exercises done above 70% of 1RM a velocity loss of 40% is considered the point of failure, eg. RPE-10. A velocity loss of only 10–15% across the set is closer to an RPE of 6–7. The table below gives you a good guide to how velocity loss matches up with RPE.

This table shows the approximate relationship between velocity loss and RPE based on current research consensus.

Some have suggested that velocity loss should be used a replacement for RPE but I strongly disagree. The magic happens when you combine the objective data of velocity tracking with the lifter's internal *perceived*

exertion assessment.

I use velocity loss in combination with RPE in two ways.

The simplest is to compare velocity loss *after* you have already noted the set RPE. Doing this helps me course correct for my next set, and helps sharpen my assessment of RPE over time. Any discrepancy between my own RPE and the velocity loss from a set gives me the chance to fine tune my internal RPE scoring system, and may alert me that I am not as recovered as I think I am.

Metric allows you to track both velocity and RPE so you can do both of these things directly in the app.

The second method is to end sets at a fixed velocity loss point. A program may plan for a set of 90kg for 6 reps *or* when you hit 25% velocity loss, whichever comes first. This is a bit harder to implement as you need to monitor live intra-set velocity data (typically with a coach watching). Our roadmap for Metric includes a tool to do this automatically for you!

5. Use your minimum velocity threshold to estimate 1RM

Knowing the velocity at which your 1RM occurs can be a valuable piece of training data to program with or track progress. This is known as the *minimum velocity threshold*, and is the mean rep velocity below which you fail to continue pushing (or pulling) the bar. Importantly this number differs for each lift (and lifter), so you can't take a squat minimum velocity threshold and use it for a bench press!

The table below shows researched minimum velocity thresholds for

the main lifts. These can serve as a good starting point for you to work from. If you want to know your precise individual minimum velocity threshold you will need to attempt an actual 1RM (with proper safety equipment and supervision), and record the velocity for that.

The table shows the minimum velocity thresholds for several lifts typically found in research.

With a known minimum velocity threshold for each lift, you can derive your estimated 1RM (e1RM) at any time using sub-maximal sets. Collect the best rep velocity for a series of progressively heavier sets on a given exercise to generate a *load-velocity profile*. A load velocity profile is just a linear trend line fit to your data - every spreadsheet or charting app can do that for you, no maths required!

In the Metric app you can set a custom MVT value for each of your exercises and this will be used to find your [e1RM from your velocity tracking data in every workout](#) automatically.

Where the load-velocity profile hits your minimum velocity threshold is your e1RM.

This
graph
shows
some of
my
recent
training
data.
Even
though I
only
have
four
samples
(50, 70,
90, 120kg
lifts), I
can
estimate
my 1RM
is 175kg.

Because this system only requires a few sub-maximal sets, you can calculate your e1RM every single session just by tracking your warm up sets — no need to do highly taxing maximal sets!

I like to use this method to track progress over the course of a training block. Ideally I want to see my e1RM increase most sessions as I hit faster velocities across my warm up sets.

A more advanced option is to use e1RMs as an alternative to percentage 1RM based programming. There is [some evidence](#) that programming with e1RM derived from velocity data produces better adaptations compared to calculating percentages from 1RM tests done prior to the training block.

The value of your e1RM will reflect your readiness on that specific day

and the load and rep schemes should therefore auto-regulate themselves to reflect an appropriate training stimulus. This use of e1RM or velocity is a bit convoluted and can be an overly sensitive system for novice lifters, but if you are accustomed to programming with percentage systems it can be a good first step to integrate velocity into your training plan.

Find what works best for you

Velocity is a fantastic tool for optimising your training sessions in real-time in a variety of ways. There are many ways to include VBT into your existing training flow, but don't feel like you have to use them all!

If you are starting out with VBT, keep it simple and use velocity in a way that fits in your goals or training style: motivate your athletes and increase their intent to move with competition or a leaderboard, track changes over time with velocity PRs, or e1RM profiling and use warm-up velocities to adjust work set loads, dial in your range of motion performance, or set up velocity loss cut-off points.

There is no right or wrong, and you don't have to use all of these approaches. I recommend you experiment and cherry-pick the methods that align best with your coaching or training goals.

If you haven't already, make sure to download my free logbook on the [VBTcoach website](#), and come say hi on the [Metric VBT instagram channel](#) if you have any questions about how to best implement VBT.

Happy lifting!

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